

Bayesian Optimization Based on Multi-Fidelity Data

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Formal Problem Statement

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The design parameters are $\mathbf{x} \in [0, 1]^p$, and the simulator is a black box $y = h(\mathbf{x})$. There are two evaluation channels, with cost and accuracy layered as follows:

- Low-fidelity $h_L(\mathbf{x})$: cost c_L , with systematic bias;
- High-fidelity $h_H(\mathbf{x})$: cost $c_H \gg c_L$, accurate.

There is a bias structure between the two fidelities, whose form is unknown. The total budget is 15 high-fidelity equivalents, and we define $\hat{h} = h_H(\hat{\mathbf{x}})$. Task: estimate as accurately as possible

$$\hat{\mathbf{x}} \approx \arg \min_{\mathbf{x} \in [0, 1]^p} h_H(\mathbf{x})$$

Theoretical Foundations

Maximum Projection Design

Maximin Latin hypercube sampling design (Morris and Mitchell, 1995)[1]:

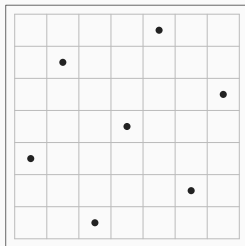


Figure 1: Maximin Latin hypercube sampling design

Maximum Projection Design

Maximum projection design (Joseph et al., 2015)[2]: let $D = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ be a design with n points on $[0, 1]^p$, whose criterion is

$$\min_D \sum_{i=1}^n \prod_{j \neq i} \frac{1}{\sum_{k=1}^p (x_{ik} - x_{jk})^2} \quad (1)$$

Maximum Projection Design

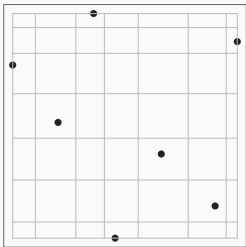


Figure 2: Maximum projection design

Maximum OFAT Design with Projection

Estimator of the total Sobol' index V_i^{tot} (Jansen, 1999)[3]:

$$\widehat{V_i^{\text{tot}}} = \frac{1}{2m} \left\| h(\mathbf{A}) - h(\mathbf{A}^{(\mathbf{B}_i)}) \right\|^2 \quad (2)$$

where $\mathbf{A}^{(\mathbf{B}_i)}$ is the matrix obtained by replacing the i -th column of the $m \times p$ matrix $\mathbf{A}$ with the i -th column of $\mathbf{B}$, $\|\cdot\|$ is the Euclidean norm, and $h(\mathbf{x})$ is an arbitrary function to be tested.

Expectation of the total Sobol' index V_i^{tot} :

$$\mathbb{E} \left(\widehat{V_i^{\text{tot}}} \right) = \frac{\tau^2 \sqrt{w_i}}{2m} \sum_{j=1}^m \left| \mathbf{A}_{ji} - \mathbf{A}_{ji}^{(i)} \right| \quad (3)$$

Maximum OFAT Design with Projection

Criterion for maximizing the total Sobol' index V_i^{tot} :

$$\arg \max_{\mathbf{A}} \sum_{j=1}^m \left| \mathbf{A}_{ji} - \mathbf{A}_{ji}^{(i)} \right| \quad (4)$$

Equation (4) is called the **maximum one-factor-at-a-time design**.

Maximum OFAT Design with Projection

Projection mapping:

$$x_i \leftarrow \begin{cases} x_i - \frac{0.25}{m}, & x_i < 0.5 \\ x_i + \frac{0.25}{m}, & x_i \geq 0.5 \\ x_i, & \text{otherwise} \end{cases} \quad , \quad x_i \leftarrow \begin{cases} x_i + \frac{0.25}{m}, & x_i < 0.5 \\ x_i - \frac{0.25}{m}, & x_i \geq 0.5 \\ x_i, & \text{otherwise} \end{cases} \quad (5)$$

Applying (4) after the mapping in (5) gives the **maximum one-factor-at-a-time design with projection**.

Maximum OFAT Design with Projection

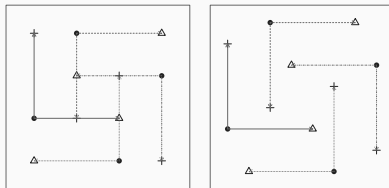


Figure 3: Maximum one-factor-at-a-time design (left) and maximum one-factor-at-a-time design with projection (right)

Sequential design: let $\mathcal{X} \subset \mathbb{R}^p$ be the design space; first choose $x_1 \in \mathcal{X}$, then add $x_2, \dots, x_n$ one by one as follows (Joseph, 2016)[4]:

$$\mathbf{x}_{k+1} = \arg \min_{\mathbf{x} \in \mathcal{X}} \sum_{i=1}^k \frac{1}{\prod_{j=1}^p (x_j - x_{ij})^2} \quad (6)$$

where $k = 1, \dots, n - 1$.

Nested Design

Suppose there are K fidelity levels, where “1” denotes the lowest fidelity and “ K ” the highest fidelity, and the corresponding sets of design points are $\{D_i\}_{i=1}^K$.

Fusion formula (Kennedy and O’Hagan, 2000)[5]:

$$h_k(\mathbf{x}) = h_{k-1}(\mathbf{x}) + \delta_k(\mathbf{x}) \quad (7)$$

where $k = 2, \dots, K$, and $\delta_k(\mathbf{x})$ denotes the bias between the outputs of the $(k-1)$ -th and the k -th fidelity levels. We make the following Gaussian process assumptions:

$$\begin{aligned} \delta_k(\mathbf{x}) &\sim \mathcal{GP}\{0, \tau_k^2 R_k(\cdot)\} \\ h_L(\mathbf{x}) &\sim \mathcal{GP}\{\mu, \tau_L^2 R_L(\cdot)\} \end{aligned} \quad (8)$$

Definition (Gaussian Process)

If for arbitrarily chosen $\mathbf{x}_1, \dots, \mathbf{x}_n$ and any integer $n \geq 1$, the vector $\mathbf{Y} = \{Y(\mathbf{x}_1), \dots, Y(\mathbf{x}_n)\}'$ follows a multivariate normal distribution, then the random function $Y(\mathbf{x})$ is called a Gaussian process.

Nested Design

Nested design (Kennedy and O'Hagan, 2000): $D_1 \supset D_2 \supset \dots \supset D_K$.

Sequential maximum-entropy (Shannon, 1948)[6] design problem:

$$\begin{aligned} D_1^* &= \arg \max_{D_1} |\mathbf{R}_1(D_1, D_1)| \\ D_2^* &= \arg \max_{D_2 \subseteq D_1^*} |\mathbf{R}_2(D_2, D_2)| \end{aligned} \tag{9}$$

where $|\cdot|$ is the determinant, and $\mathbf{R}_k(D, D')$ is the correlation matrix of the k -th fidelity level whose (i, j) element is $R_k(\mathbf{a}_i - \mathbf{b}_j)$.

Nested Design

The optimal sample point to remove from D_1 is (Joseph, 2026)[7]:

$$x_k^* = \arg \max_{x_k \in D_1} \sum_{j \neq k} \frac{1}{\prod_{l=1}^p \{|x_{kl} - x_{jl}| + \beta_l\}^2} \quad (10)$$

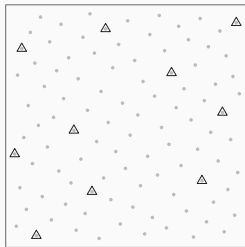


Figure 4: Nested design with two fidelity levels

Expected Improvement Method

Optimization objective of the black-box function:

$$\min_{\mathbf{x} \in [0,1]^p} h(\mathbf{x}) \quad (11)$$

Iterative formula for finding the next design point:

$$\mathbf{x}_{n+1} = \arg \min_{\mathbf{x}} \widehat{h}_n(\mathbf{x}) \quad (12)$$

This procedure is repeated until convergence. Since the Gaussian process adopts a Bayesian framework, such problems are called **Bayesian optimization methods**. Expected improvement method (Jones et al., 1998)[8]: let the experimental design be $D_n = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, with observed data $\mathbf{y}_n = (y_1, \dots, y_n)'$.

Expected Improvement Method

Place a Gaussian process prior on the unknown function:

$$h(\mathbf{x}) \sim \mathcal{GP}\{\mu, \tau^2 R(\cdot)\} \quad (13)$$

where μ is the constant mean, τ^2 the variance, and $R(\cdot)$ the correlation function. Let $\mathbf{R}_n$ be the $n \times n$ correlation matrix whose (i, j) element is $R(\mathbf{x}_i - \mathbf{x}_j)$, $\mathbf{r}_n(\mathbf{x})$ the correlation vector whose i -th element is $R(\mathbf{x} - \mathbf{x}_i)$, and $\mathbf{1}_n$ the n -dimensional column vector of all ones.

Expected Improvement Method

Its posterior distribution (Joseph, 2026, Lemma 2.2):

$$h(\mathbf{x}) \mid D_n, \mathbf{y}_n \sim \mathcal{N} \left\{ \mu + \mathbf{r}_n(\mathbf{x})' \mathbf{R}_n^{-1} (\mathbf{y}_n - \mu \mathbf{1}_n), \right. \\ \left. \tau^2 \left(1 - \mathbf{r}_n\{\mathbf{x}\}' \mathbf{R}_n^{-1} \mathbf{r}_n\{\mathbf{x}\} \right) \right\} \quad (14)$$

Let $h_{\min}^{(n)} = \min \mathbf{y}_n$, and define the improvement function:

$$I(\mathbf{x}) = \max \left\{ h_{\min}^{(n)} - h(\mathbf{x}), 0 \right\} \quad (15)$$

Thus, if the chosen x satisfies $h(\mathbf{x}) < h_{\min}^{(n)}$, an **improvement is obtained**; otherwise there is no improvement.

Expected Improvement Method

Iterative formula for expected improvement:

$$\mathbf{x}_{n+1} = \arg \max_{\mathbf{x}} \mathbb{E} \{ I(\mathbf{x}) \mid D_n, \mathbf{y}_n \} \quad (16)$$

Explicit expression of expected improvement (Jones et al., 1998):

$$\mathbf{x}_{n+1} = \arg \max_{\mathbf{x}} \{ s_n(\mathbf{x}) [u_n\{\mathbf{x}\} \Phi(u_n\{\mathbf{x}\}) + \phi(u_n\{\mathbf{x}\})] \} \quad (17)$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ denote the standard normal probability density function and the standard normal cumulative distribution function, respectively; $s_n(\mathbf{x})$ is the posterior standard deviation, and $u_n\{\mathbf{x}\} = \{ h_{\min}^{(n)} - \widehat{h}_n(\mathbf{x}) \} / s_n(\mathbf{x})$.

Expected Improvement Method

Derivatives of the explicit expression:

$$\frac{\partial \text{EI}}{\partial \widehat{h}_n(\mathbf{x})} = -\Phi\{u_n\{\mathbf{x}\}\} < 0, \quad \frac{\partial \text{EI}}{\partial s_n(\mathbf{x})} = \phi\{u_n\{\mathbf{x}\}\} > 0 \quad (18)$$

where $\text{EI} = s_n(\mathbf{x}) [u_n\{\mathbf{x}\} \Phi(u_n\{\mathbf{x}\}) + \phi(u_n\{\mathbf{x}\})]$

Expected Improvement Method

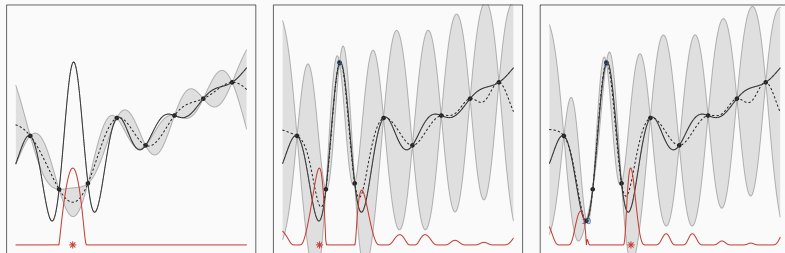


Figure 5: Two iterations of the expected improvement minimization algorithm on a test function

One-dimensional test function (Joseph, 2026):

$$h(x) = \frac{\sin(10\pi x)}{1 + 64(x - 0.25)^2} + x^2 \quad (19)$$

where $x \in [0, 1]$.

Two-dimensional Branin function (Surjanovic and Bingham, 2013)[9]:

$$f(\mathbf{x}) = \left(x_2 - \frac{5.1}{4\pi^2}x_1^2 + \frac{5}{\pi}x_1 - 6\right)^2 + 10\left(1 - \frac{1}{8\pi}\right)\cos x_1 + 10 \quad (20)$$

where $x_1 \in [-5, 10]$ and $x_2 \in [0, 15]$.

Test Functions

Simple additive function (Joseph, 2026):

$$h(\mathbf{x}) = \sum_{k=1}^4 \frac{1}{k} \sin(\{k^2 + 1\} \pi x_k) \quad (21)$$

where $\mathbf{x} \in [0, 1]^4$.

Function simulating borehole water flow (Morris et al., 1993)[10]:

$$y = \frac{2\pi x_3 (x_4 - x_6)}{\log\left(\frac{x_2}{x_1}\right) \left\{1 + \frac{2x_7 x_3}{\log(x_2/x_1) x_1^2 x_8} + \frac{x_3}{x_5}\right\}} \quad (22)$$

where $x_1 \in [0.05, 0.15]$, $x_2 \in [100, 50000]$, $x_3 \in [63070, 115600]$,
 $x_4 \in [990, 1110]$, $x_5 \in [63.1, 116]$, $x_6 \in [700, 820]$, $x_7 \in [1120, 1680]$,
 $x_8 \in [9855, 12045]$.

General formula for constructing multi-fidelity functions (Song et al., 2024)[11]:

$$f_{\text{L}}(\mathbf{x}) = f_{\text{H}}(\mathbf{x}) + \frac{c}{p} \cdot \sum_{i=1}^p \left(x_i - \frac{1}{2} \right) \quad (23)$$

Low-fidelity versions of the four test functions:

$$f_L(x) = f_H(x) + 0.3 \left(x - \frac{1}{2} \right) \quad (24)$$

$$f_L(\mathbf{u}) = f_H(\mathbf{u}) + 40 \cdot \frac{u_1 + u_2 - 1}{2} \quad (25)$$

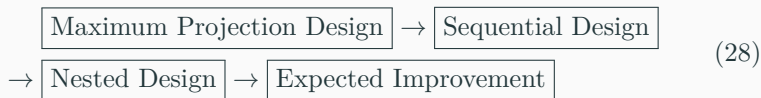
$$f_L(\mathbf{x}) = f_H(\mathbf{x}) + 0.5 \cdot \frac{\sum_{k=1}^4 (x_k - \frac{1}{2})}{4} \quad (26)$$

$$f_L(\mathbf{u}) = f_H(\mathbf{u}) + 70 \cdot \frac{\sum_{i=1}^8 (u_i - \frac{1}{2})}{8} \quad (27)$$

Complete Pipeline

Complete Pipeline

Complete pipeline M1:



Blank control M0:



Complete Pipeline

Removing only the nested design S1:



Removing only the sequential design S2:



Calibration Experiment

Output 1: optimal number of high-fidelity evaluations for each dimension

n2* by dimension: d1=2 d2=14 d4=14 d8=14

Following the M1 workflow, the room for the sequential and fusion components is squeezed out almost entirely by the budget.

Ablation Experiment

Output 2: table of function minima fitted by the four experimental configurations in each of the four dimensions after forcing 5 high-fidelity evaluations

config	d1	d2	d4	d8
M1	-0.6035453	3.1626952	-1.410891	19.68542
S2	-0.6035453	0.4041193	-1.606024	12.70143
S1	0.0000000	5.4693989	-1.187530	17.15853
M0	0.0000000	0.8787899	-1.297524	10.50205
M0@n2=14	-0.5776889	2.2567621	-1.448456	12.59210

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- In the 1-dimensional case, the expected improvement method **fails**.
- In the 4-dimensional case, the fusion model uses a **large amount of low-fidelity data** to correct the approximation of the high-fidelity surface.
- The screening component provides **no positive contribution** in any of the four dimensions: the bias structure between low and high fidelity makes the factor importance ranking unreliable.

Ablation Experiment

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- The value of fusion decays as the dimension increases: in the 8-dimensional case, the blank control is actually better than the fusion-only configuration, which is a manifestation of the **curse of dimensionality** within the multi-fidelity framework.
- Under the same budget, a few high-fidelity points combined with fusion **far outperform** running with many high-fidelity points alone.

Summary

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- Under an extremely small budget, the initial design **dominates the results**; the room for the sequential and fusion components is squeezed out almost entirely by the budget.
- Fusion is the **decisive component**: in low dimensions, it goes from failure to the global optimum; in high dimensions, it is constrained by the curse of dimensionality.
- The screening component provides **no positive contribution**: the bias structure between low and high fidelity makes the factor importance ranking unreliable.
- Under the same budget, a few high-fidelity points combined with fusion **far outperform** running with many high-fidelity points alone.

Future Work

Direction for improvement: under what bound on the total Sobol' index of the bias field $\delta(\mathbf{x})$ does the factor importance ranking given by the low-fidelity function preserve the true high-fidelity ranking?

All the above codes, figures and tex source files have been uploaded to

<https://github.com/wenje-ma/singapore>

Everyone is welcome to reproduce the results.

Thank you for listening!

References

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